

ASSIGNMENT - 1

1. The radius of a wheel is 14 cm. The disc rolls $\left(\frac{3}{4}\right)^{th}$ of a circumference. Find the distance moved by the centre of wheel.

Sol Given,

$$r = 14 \text{ cm}$$

Ground distance moved

$$\begin{aligned} &= \frac{3}{4} \times 2\pi r \\ &= \frac{3}{2} \times \frac{22}{7} \times 14 \\ &= 66 \text{ cm} \end{aligned}$$

Therefore, the centre of the wheel moves 66 cm.

2. The minute hand of a watch is 2 cm long. How far its tip will move in 10 min? (Take $\pi = 3.14$)

Sol In 1 min, MH moves 6°

In 10 min, MH moves 60°

So, angle traversed in 10 min,

$$\therefore \theta = 60^\circ = \left(60 \times \frac{\pi}{180}\right) = \left(\frac{\pi}{3}\right)^c$$

Hence, the distance moved by the tip of the minute hand,

$$l = r\theta$$

$$= 2 \times \frac{\pi}{3}$$

$$\approx 2.1 \text{ cm}$$

3. Which of the following statements holds good if $m = \tan 3$ and $n = \sec 6$?

(a) m and n both are positive

(b) m and n both are negative

(c) m is positive and n is negative

(d) m is negative and n is positive

Sol

$$m = \tan 3$$

$$3 \in \left(\frac{\pi}{2}, \pi \right) \Rightarrow \tan 3 < 0 \quad \left(\because \theta \in \left(\frac{\pi}{2}, \pi \right) \Rightarrow \tan \theta < 0 \right)$$

$$\therefore m < 0$$

$$n = \sec 6$$

$$6 \in \left(\frac{3\pi}{2}, 2\pi \right) \Rightarrow \sec 6 > 0 \quad \left(\because \theta \in \left(\frac{3\pi}{2}, 2\pi \right) \Rightarrow \sec \theta > 0 \right)$$

$$\therefore n > 0$$

4. In any quadrilateral ABCD, show that $\cos A \cos B - \cos C \cos D = \sin A \sin B - \sin C \sin D$.

Sol Since ABCD is a quadrilateral, $A + B + C + D = 2\pi$

$$\Rightarrow A + B = 2\pi - (C + D)$$

$$\Rightarrow \cos(A + B) = \cos(2\pi - (C + D))$$

$$\Rightarrow \cos(A + B) = \cos(C + D)$$

$$\Rightarrow \cos A \cos B - \sin A \sin B = \cos C \cos D - \sin C \sin D$$

$$\Rightarrow \cos A \cos B - \cos C \cos D = \sin A \sin B - \sin C \sin D$$

5. Prove that $\frac{\cos 2\theta}{1 + \sin 2\theta} = \tan\left(\frac{\pi}{4} - \theta\right)$.

$$\text{LHS} = \frac{\cos 2\theta}{1 + \sin 2\theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta}$$

$$= \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{(\cos \theta + \sin \theta)^2}$$

Sol

$$= \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$$

$$= \tan\left(\frac{\pi}{4} - \theta\right)$$

$$= \text{RHS}$$

6. $\cos 12^\circ \cos 24^\circ \cos 48^\circ \cos 84^\circ =$

(a) $\frac{1}{8}$

(b) $\frac{1}{16}$

(c) $\frac{1}{32}$

(d) None of these

Sol

$$\begin{aligned}
& \cos 12^\circ \cos 24^\circ \cos 48^\circ \cos 84^\circ \\
&= \cos(12^\circ) \cos(2 \times 12^\circ) \cos(2^2 \times 12^\circ) \cos(90^\circ - 6^\circ) \\
&= \frac{\sin(2^3 \times 12^\circ)}{2^3 \sin 12^\circ} \times \sin 6^\circ \\
&= \frac{\sin 96^\circ \sin 6^\circ}{8 \sin 12^\circ} \\
&= \frac{\sin(90^\circ + 6^\circ) \sin 6^\circ}{16 \sin 6^\circ \cos 6^\circ} \\
&= \frac{\cos 6^\circ}{16 \cos 6^\circ} \\
&= \frac{1}{16}
\end{aligned}$$

7. $\sin \frac{8\pi}{18} \sin \frac{7\pi}{18} \sin \frac{5\pi}{18} \sin \frac{\pi}{18} =$

(a) $\cos \frac{\pi}{18}$

(b) $\frac{1}{8} \cos \frac{\pi}{18}$

(c) $\frac{1}{2} \cos \frac{\pi}{18}$

(d) $\frac{1}{16} \cos \frac{\pi}{18}$

Sol

$$\begin{aligned}
& \sin \frac{8\pi}{18} \sin \frac{7\pi}{18} \sin \frac{5\pi}{18} \sin \frac{\pi}{18} \\
&= \cos\left(\frac{\pi}{2} - \frac{8\pi}{18}\right) \cos\left(\frac{\pi}{2} - \frac{7\pi}{18}\right) \cos\left(\frac{\pi}{2} - \frac{5\pi}{18}\right) \cos\left(\frac{\pi}{2} - \frac{\pi}{18}\right) \\
&= \cos \frac{\pi}{18} \cos \frac{2\pi}{18} \cos \frac{4\pi}{18} \cos \frac{8\pi}{18} \\
&= \frac{\sin\left(2^4 \times \frac{\pi}{18}\right)}{2^4 \sin \frac{\pi}{18}} \\
&= \frac{\sin \frac{16\pi}{18}}{16 \sin \frac{\pi}{18}} \\
&= \frac{\sin\left(\pi - \frac{16\pi}{18}\right)}{16 \sin \frac{\pi}{18}} \\
&= \frac{\sin \frac{2\pi}{18}}{16 \sin \frac{\pi}{18}} \\
&= \frac{2 \sin \frac{\pi}{18} \cos \frac{\pi}{18}}{16 \sin \frac{\pi}{18}} \\
&= \frac{1}{8} \cos \frac{\pi}{18}
\end{aligned}$$

8. Prove that $(\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right)$.

Sol

$$\begin{aligned} \text{LHS} &= (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 \\ &= \cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta + \sin^2 \alpha + \sin^2 \beta + 2 \sin \alpha \sin \beta \\ &= 1 + 2 \cos \alpha \cos \beta + 1 + 2 \sin \alpha \sin \beta \\ &= 2 + 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ &= 2 + 2 \cos(\alpha - \beta) \\ &= 2(1 + \cos(\alpha - \beta)) \\ &= 2 \left(2 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \right) \\ &= 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \\ &= \text{RHS} \end{aligned}$$

9. Prove that $\sin 4A = 4 \sin A \cos^3 A - 4 \cos A \sin^3 A$

Sol

$$\begin{aligned} \text{LHS} &= \sin 4A \\ &= 2 \sin 2A \cos 2A \\ &= 2(2 \sin A \cos A)(\cos^2 A - \sin^2 A) \\ &= 4 \sin A \cos^3 A - 4 \cos A \sin^3 A \\ &= \text{RHS} \end{aligned}$$

10. (i) If $\cos x = -\frac{3}{5}$ and x lies in the IIIrd quadrant, then find the values of $\cos \frac{x}{2}, \sin \frac{x}{2}, \sin 2x$.

- (ii) If $\cos x = -\frac{3}{5}$ and x lies in the IInd quadrant, then find the values of $\sin 2x$ and $\sin \frac{x}{2}$.

Sol (i) $\cos x = -\frac{3}{5}, x \in \left(\pi, \frac{3\pi}{2} \right)$

$$x \in \left(\pi, \frac{3\pi}{2} \right) \Rightarrow \frac{x}{2} \in \left(\frac{\pi}{2}, \frac{3\pi}{4} \right)$$

$$\therefore \cos \frac{x}{2} < 0, \sin \frac{x}{2} > 0$$

$$\begin{aligned}\cos \frac{x}{2} &= -\sqrt{\frac{1+\cos x}{2}} \\&= -\sqrt{\frac{1+\frac{3}{5}}{2}} \\&= -\sqrt{\frac{2}{10}} \\&= -\frac{1}{\sqrt{5}}\end{aligned}$$

$$\begin{aligned}\sin \frac{x}{2} &= \sqrt{\frac{1-\cos x}{2}} \\&= \sqrt{\frac{1-\frac{3}{5}}{2}} \\&= \sqrt{\frac{8}{10}} \\&= \sqrt{\frac{4}{5}} \\&= \frac{2}{\sqrt{5}}\end{aligned}$$

$$\begin{aligned}\sin 2x &= 2 \sin x \cos x \\&= 2 \times \left(-\sqrt{1-\left(\frac{-3}{5}\right)^2} \right) \times \left(\frac{-3}{5} \right) \\&= 2 \times \frac{4}{5} \times \frac{3}{5} \\&= \frac{24}{25}\end{aligned}$$

$$(ii) \quad \cos x = \frac{-3}{5}, x \in \left(\frac{\pi}{2}, \pi \right)$$

$$\sin 2x = 2 \sin x \cos x$$

$$\begin{aligned} &= 2 \times \sqrt{1 - \left(\frac{-3}{5}\right)^2} \times \left(\frac{-3}{5}\right) \\ &= -2 \times \frac{4}{5} \times \frac{3}{5} \\ &= \frac{-24}{25} \end{aligned}$$

$$x \in \left(\frac{\pi}{2}, \pi\right) \Rightarrow \frac{x}{2} \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\therefore \sin \frac{x}{2} > 0$$

$$\begin{aligned} \sin \frac{x}{2} &= \sqrt{\frac{1 - \cos x}{2}} \\ &= \sqrt{\frac{1 - \left(\frac{-3}{5}\right)}{2}} \\ &= \sqrt{\frac{8}{10}} \\ &= \sqrt{\frac{4}{5}} \\ &= \frac{2}{\sqrt{5}} \end{aligned}$$

11. If $\tan(2\alpha + 3\theta) = n \tan(2\alpha - 3\theta)$, then show that $(n+1)\sin 6\theta = (n-1)\sin 4\alpha$.

Sol

$$\tan(2\alpha + 3\theta) = n \tan(2\alpha - 3\theta)$$

$$\Rightarrow n = \frac{\tan(2\alpha + 3\theta)}{\tan(2\alpha - 3\theta)}$$

Using componendo and dividendo, we have,

$$\begin{aligned} \Rightarrow \frac{n+1}{n-1} &= \frac{\tan(2\alpha + 3\theta) + \tan(2\alpha - 3\theta)}{\tan(2\alpha + 3\theta) - \tan(2\alpha - 3\theta)} \\ &= \frac{\frac{\sin(2\alpha + 3\theta)}{\cos(2\alpha + 3\theta)} + \frac{\sin(2\alpha - 3\theta)}{\cos(2\alpha - 3\theta)}}{\frac{\sin(2\alpha + 3\theta)}{\cos(2\alpha + 3\theta)} - \frac{\sin(2\alpha - 3\theta)}{\cos(2\alpha - 3\theta)}} \\ &= \frac{\frac{\sin(2\alpha + 3\theta)\cos(2\alpha - 3\theta) + \sin(2\alpha - 3\theta)\cos(2\alpha + 3\theta)}{\sin(2\alpha + 3\theta)\cos(2\alpha - 3\theta) - \sin(2\alpha - 3\theta)\cos(2\alpha + 3\theta)}}{\frac{\sin(2\alpha + 3\theta + 2\alpha - 3\theta)}{\sin(2\alpha + 3\theta - 2\alpha + 3\theta)}} \\ &= \frac{\sin 4\alpha}{\sin 6\theta} \end{aligned}$$

$$\therefore (n+1)\sin 6\theta = (n-1)\sin 4\alpha$$

12. If $0 < x < \frac{\pi}{4}$ and $\cos x + \sin x = \frac{5}{4}$, then find the numerical values of $\cos x - \sin x$.

Sol

$$\cos x + \sin x = \frac{5}{4}, x \in \left(0, \frac{\pi}{4}\right)$$

$$\cos x - \sin x = k$$

$$\therefore (\cos x + \sin x)^2 + (\cos x - \sin x)^2 = \left(\frac{5}{4}\right)^2 + k^2$$

$$\Rightarrow \cos^2 x + \sin^2 x + 2 \cos x \sin x + \cos^2 x + \sin^2 x - 2 \cos x \sin x = \frac{25}{16} + k^2$$

$$\Rightarrow 2 = \frac{25}{16} + k^2$$

$$\Rightarrow k^2 = 2 - \frac{25}{16}$$

$$\Rightarrow k^2 = \frac{7}{16}$$

$$\Rightarrow k = \frac{\pm\sqrt{7}}{4}$$

$$\therefore \cos x - \sin x = \frac{\pm\sqrt{7}}{4}$$

13. Prove that: $\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2 \left(\frac{3\pi}{4} + x \right)$

Sol

$$\begin{aligned}\text{LHS} &= \frac{1 + \sin 2x}{1 - \sin 2x} \\&= \frac{1 + \frac{2 \tan x}{1 + \tan^2 x}}{1 - \frac{2 \tan x}{1 + \tan^2 x}} \\&= \frac{1 + \tan^2 x + 2 \tan x}{1 + \tan^2 x - 2 \tan x} \\&= \frac{(1 + \tan x)^2}{(1 - \tan x)^2} \\&= \left(\frac{1 + \tan x}{1 - \tan x} \right)^2 \\&= \tan^2 \left(\frac{\pi}{4} + x \right) \\&= \cot^2 \left(\frac{\pi}{2} + \frac{\pi}{4} + x \right) \\&= \cot^2 \left(\frac{3\pi}{4} + x \right) \\&= \text{RHS}\end{aligned}$$

14. Prove the identity $\left(\frac{\sin 3\theta}{\sin \theta} \right)^2 - \left(\frac{\cos 3\theta}{\cos \theta} \right)^2 = 8 \cos 2\theta$, wherever it is defined.

Sol

$$\begin{aligned}
\text{LHS} &= \left(\frac{\sin 3\theta}{\sin \theta} \right)^2 - \left(\frac{\cos 3\theta}{\cos \theta} \right)^2 \\
&= \left(\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} \right) \left(\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} \right) \\
&= \frac{(\sin 3\theta \cos \theta + \cos 3\theta \sin \theta)(\sin 3\theta \cos \theta - \cos 3\theta \sin \theta)}{\sin^2 \theta \cos^2 \theta} \\
&= \frac{\sin(3\theta + \theta) \sin(3\theta - \theta)}{(\sin \theta \cos \theta)^2} \\
&= \frac{\sin 4\theta \sin 2\theta}{\left(\frac{\sin 2\theta}{2} \right)^2} \\
&= \frac{4 \sin 4\theta \sin 2\theta}{\sin^2 2\theta} \\
&= \frac{4 \sin 4\theta}{\sin 2\theta} \\
&= \frac{8 \sin 2\theta \cos 2\theta}{\sin 2\theta} \\
&= 8 \cos 2\theta \\
&= \text{RHS}
\end{aligned}$$

15. Express the following in terms of sine only

(i) $3\sin x + 4\cos x$

(ii) $\cos x + \sin x$

(iii) $\cos x - \sin x$

$$3\sin x + 4\cos x = 5 \left(\frac{3}{5} \sin x + \frac{4}{5} \cos x \right)$$

Sol (i) $= 5(\cos \phi \sin x + \sin \phi \cos x)$
 $= 5 \sin(x + \phi), \text{ where } \tan \phi = \frac{4}{3}$

(ii) $\cos x + \sin x = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right)$
 $= \sqrt{2} \left(\sin \frac{\pi}{4} \cos x + \cos \frac{\pi}{4} \sin x \right)$
 $= \sqrt{2} \sin \left(\frac{\pi}{4} + x \right)$

$$\begin{aligned}\cos x - \sin x &= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right) \\ \text{(iii)} \quad &= \sqrt{2} \left(\sin \frac{\pi}{4} \cos x - \cos \frac{\pi}{4} \sin x \right) \\ &= \sqrt{2} \sin \left(\frac{\pi}{4} - x \right)\end{aligned}$$

